Ashirbad Mishra

Doctoral Fellow
Dept. of Environmental Science,
Berhampur University
Bhanja Bihar-760007, Odisha, India

Mob: +91-7064560805
Email: am.rs.evs@buodisha.edu.in
ashirbadmishra10@gmail.com
Google Scholar, LinkedIn, ORCID, ResearchGate
https://ashirbad-mishra.github.io/

Education:

2024 - Present	**Doctoral Fellow** at Dept. of Environmental Science, Berhampur University, India **Thesis:** Development of High Resolution Particulate & Indirect GHGs Emission and its Teleconnection with Disease Prevalence across India **Subjects Studied:** Air pollution, Emission Inventory, Urban Air Quality, Public Health
2019 - 2021	**Master of Science** at Dept of Environmental Science, Fakir Mohan University, India **Thesis:** Potential of Carbon dioxide sequestration methodologies, a brief review **Subjects Studied:** Pollution Management, Remote Sensing & Geographic Information System, Environmental impact assessment, Disaster management
2015 - 2019	**Bachelor of Technology** at Dept of Civil & Environmental Engineering, Veer Surendra Sai University of Technology, India **Thesis:** Effect of confinement pressure on compressive strength of cylindrical concrete column using fly ash mixed recycled aggregate concrete **Subjects Studied:** Environmental Engineering, Pollution Measurement & Management

Research Experience:

Mar 2024 - Present — **Doctoral Fellow** at Dept. of Environmental Science, Berhampur University, India

Nature of Work:

i. Developing comprehensive high-resolution gridded emission inventories for Greenhouse Gases (GHGs) and particulate matter across India to identify pollution hotspots and determine critical sectors to target.

ii. Quantifying the relationship between poor air quality and disease prevalence, generating data-driven insights on how air pollution exposure affects public health outcomes.

May 2022 - Feb 2024 — **Research Fellow** at Dept. of Environmental Science, Utkal University, India

Nature of Work:

i. Developed spatially resolved emission inventories and analysed observational datasets to support investigations of pollutant transport and the influence of South Asian emissions on regional air quality and climate interactions.

Research Metrics: (Updated October 2026)

Peer-reviewed publications: **12**	First-author publications: **02**	Conference Presentations: **18**
Total citations: **139**	h-index: **7**	i10-index: **6**

Research Interests:

- Greenhouse Gas Emission Inventory
- Air Pollution Exposure and Environmental Health
- Atmospheric Modelling
- Urban Air quality and Source Attribution
- Satellite Remote Sensing
- Emission Mitigation Strategies and Policy Implications

List of Publications:

1. **Mishra, A.**, Mangaraj, P., Mahapatra, D., Sahoo, P., Beig, G., Panigrahi, G. R., Pattnayak, K. C., Tyagi, B., Tripathy, S. K., & Sahu, S. K. (2026). Electronic tolling-FASTag technology in transport is a game changer in climate action: A sustainable policy towards Net-Zero in India. ***Transport Policy***, *183*, 104164. https://doi.org/10.1016/j.tranpol.2026.104164

2. **Mishra, A.**, Mangaraj, P., Sahoo, P., Beig, G., Janardanan, R., & Sahu, S. K. (2026). A gridded (0.1° × 0.1°) methane emission dataset for India for 2023 to redefine global climate studies. ***Earth System Science Data***, *18*(2), 1367–1378. https://doi.org/10.5194/essd-18-1367-2026

3. Mishra, M., **Mishra, A.**, Mangaraj, P., Srivastava, A. K., Beig, G., Sahoo, P., Nayak, S., Prusty, B. A. K., Choudhary, A., & Sahu, S. K. (2025). Quantification and spatial assessment of industrial Cd and Pb emission across India. ***All Earth***, *37*(1), 1–13. https://doi.org/10.1080/27669645.2025.2535845

4. Sahoo, P., **Mishra, A.**, Mangaraj, P., Ravindra, K., Beig, G., Tyagi, B., & Sahu, S. K. (2025). What drives

anthropogenic fine particulate chloride emissions in India? – A quantitative assessment of hotspots. ***The Science of the Total Environment***, *991*, 179949–179949. https://doi.org/10.1016/j.scitotenv.2025.179949

5. Sahu, S. K., **Mishra, A.**, Mangaraj, P., Yadav, R., Sahu, M. C., Beig, G., Tyagi, B., Gunthe, S. S., Kumar, A., & Mishra, M. (2025). Source-specific fine particulates emission linked to prevalence of ophthalmic cases in India. ***Scientific Reports***, *15*(1). https://doi.org/10.1038/s41598-024-82914-6
6. Sahu, S. K., Mangaraj, P., Sahoo, P., **Mishra, A.**, & Beig, G. (2024). Quantification and Spatial Analysis of Gridded ($0.1^0 \times 0.1^0$) Emission of Indirect GHGs/Air Pollutants from Anthropogenic Sources in India. ***Environmental Pollution***, *363*, 125231–125231. https://doi.org/10.1016/j.envpol.2024.125231
7. Sahoo, P., Sahu, S. K., Mangaraj, P., **Mishra, A.**, Beig, G., & Gunthe, S. S. (2024). Reporting of Gridded Ammonia Emission and Assessment of Hotspots across India: A comprehensive study of 24 anthropogenic sources. ***Journal of Hazardous Materials***, *479*, 135557–135557. https://doi.org/10.1016/j.jhazmat.2024.135557
8. Mangaraj, P., Sahu, S. K., Beig, G., **Mishra, A.**, & Sharma, S. (2024). What Makes the Indian Megacity Chennai's Air Unhealthy? - A Bottom-up Approach to Understand the Sources of Air Pollutants. ***Aerosol and Air Quality Research***, *24*(9), 240089–240089. https://doi.org/10.4209/aaqr.240089
9. Sahu, S. K., Mishra, M., **Mishra, A.**, Mangaraj, P., & Beig, G. (2024). Quantification and assessment of hazardous mercury emission from industrial process and other unattended sectors in India: A step towards mitigation. ***Journal of Hazardous Materials***, *470*, 134103–134103. https://doi.org/10.1016/j.jhazmat.2024.134103
10. Samal, A., Sahu, S. K., **Mishra, A.**, Mangaraj, P., Pani, S. K., & Beig, G. (2024). Assessment and Quantification of Methane Emission from Indian Livestock and Manure Management. ***Aerosol and Air Quality Research***, *24*(6), 230204–230204. https://doi.org/10.4209/aaqr.230204
11. Sahu, S. K., Mangaraj, P., Beig, G., Lund, M. T., Samset, B. H., Sahoo, P., & **Mishra, A.** (2023). Development and comprehensive analysis of spatially resolved technological high resolution (0.1°×0.1°) Emission Inventory of Particulate Matter for India: A step Towards Air Quality Mitigation. ***Earth System Science Data Discussion***. https://doi.org/10.5194/essd-2023-310
12. Sahu, S. K., Sahoo, P., Mangaraj, P., Beig, G., Tyagi, B., Samal, B., **Mishra, A.**, & Yadav, R. (2023). Identification and Quantification of Emission Hotspots of Air Pollutants over Bhubaneswar: A Smart City in Eastern India. ***Aerosol and Air Quality Research***, *23*(9), 230049. https://doi.org/10.4209/aaqr.230049

Research Project:

May 2022 – Oct 2022

Junior Research Fellow at the project "Quantifying the Impact of South-Asian Aerosols on Arctic and Regional Climate (QUISARC)"
Host Institution: Utkal University, India
Partner Institution: CICERO Centre for International Climate Research, Norway
Description:
A collaborative Indo-Norwegian research project aimed at quantifying the regional and Arctic climate impacts of South Asian aerosol emissions through improved emission inventories and climate analysis.
Contribution:
Developed spatially resolved emission inventories and analysed observational datasets to support investigations of South Asian aerosol emissions and their regional and Arctic climate implications. Contributed to the preparation of scientific data visualisations and technical project reports of the research findings, analytical outputs, and relevant scientific information.

Funding & Fellowships:

1. Mukhyamantri Research and Innovation Fellowship Program
   Funding Agency: Odisha State Higher Education Council, Government of Odisha
   Period: Sept 2024 – Present
   Grant amount: INR 30,000 per month
2. Junior Research Fellowship, QUISARC
   Funding Agency: Ministry of Earth Sciences, Government of India
   Period: May 2022 – Oct 2022
   Grant amount: INR 31,000 + HRA per month

Conference Participation:

Oral Presentation:

1. Development of High-Resolution Grided Methane Emission Inventory for India: A Special Focus on North Indian States; National Workshop on Climate Change; **Sindhu Central University (SCU)**, Ladakh, Oct 2026
2. Comprehensive (0.1°×0.1°) Grided Methane Emission Inventory for India: Highlighting North Indian States; National Conference on Environment, Climate and Sustainability: Science, Policy and Pathways (ECSSPP); **University of Kashmir**, Jammu & Kashmir, Sep 2026
3. Quantification of Industrial & Unregulated Mercury Emissions across India; National workshop Mercury Management in India: Bridging Science, Technology, and Policy; **IIT Madras**, Tamil Nadu, Jun 2026
4. Quantifying Emissions across the Indo-Gangetic Plain: A Multi-Pollutant Inventory; International Mini-conference Series 2025 on Short-lived Climate Pollutants (Slcps) in the Indo-gangetic Plain; **IIT Kanpur**, Uttar Pradesh, India, Aug 2025.
5. Decoding Indian Megacity Chennai's Air Quality: A Bottom-Up Analysis of Pollution Sources; International Conference on Revisiting Public Policy and Governance in Times of Climate Change- **NAPSIPAG** 2025; Berhampur University, Odisha, India, Jan 2025.
6. Mercury Emissions from Indian Industries and Unregulated Sectors: A Comprehensive Assessment; National Symposium on Aerosol Impacts to Air Quality, Human Health & Climate Change- **IASTA** 2024; Doon University, Dehradun, Uttarakhand, India, Dec 2024.
7. Mercury Emissions from Industrial Processes and Other Unattended Sectors in India: A Quantitative Study; National symposium- **TROPMET** 2024; NIT Rourkela, Odisha, India, Dec 2024.
8. Particulate Matter Emission due to vehicular driven Windblown Road dust in India; National symposium- **TROPMET** 2023; Jaipur, Rajasthan, India, Nov 2023.
9. Environmental pollution: A threat to the outbreak of Pandemic; 12th Odisha environment congress; Balasore, India, Dec 2021.

Poster Presentation:

1. Assessment of Methane Emission for Indian Agriculture and Waste Management Sector; Odisha Research Conclave; Berhampur University, Odisha, Nov 2025
2. Linkages Between Source-Specific $PM_{2.5}$ and Ophthalmic Disease Burden Across India; **IGAC-ICACGP** Early Career Researcher; virtual, Sep 2025
3. Quantifying Industrial Mercury Emissions in India: Odisha emerged as the highest emitting hotspot; Odisha Research Conclave; Fakir Mohan University, Baleshwar, Dec 2024
4. Contribution of Traffic- Induced Resuspended Road Dust to Particulate Matter Emission in India; National Conference on Environmental Challenges: Role of Science & Technology- 24th Odisha Bigyan 'O' Paribesh Congress (OBPC) - 2023; Berhampur, Odisha, India, Nov 2023.
5. Particulate Matter Emission from Transport- Induced Resuspended Road Dust in India; **IGAC- ICACGP** Early Career Researcher conference; Online, Nov 2023.
6. Establishing a link between fine particulate matter ($PM_{2.5}$) zones and COVID -19 over India based on anthropogenic emission sources and air quality data; **Royal Society of Chemistry** (**RSC**) and BCKIC Joint Symposium on Health & Wellbeing; Bhubaneswar, India, Jan 2023.
7. Development of national emission inventory and assessment of major air pollutants from open municipal solid waste (MSW) burning in India; Fall meeting 2022, **American Geophysical Union**; Virtual, Chicago, USA, Dec 2022
8. Showcasing our team's work to monitor and mitigate air pollution in India; Sumangalam Panchmahabhoot Conference series VAYU- The vital force of life; Bhubaneswar, India, Dec 2022.
9. Spatially resolved inventory for atmospheric pollutants from open MSW burning in India; National symposium- **TROPMET 2022**; IISER Bhopal, India, Dec 2022.

Professional Services:

Journal Reviewing: International Journal of Environmental Studies, Environmental Science and Pollution Research

Research Skills:

- Emission Inventory Development
- Spatial Data Analysis & Visualisation
- Satellite Data Processing
- Environmental Health

Technical Expertise:

- Spatial Data Analysis Software: ArcGIS, Google Earth Pro.

- Statistical Software: Origin, Sigma plot, SPSS
- Programming Language: PYTHON, R, and MATLAB

Collaborative Research Network:

i. University of Birmingham, UK
ii. CICERO Centre for International Climate Research, Oslo, Norway
iii. National Institute for Environmental Studies, Tsukuba, Japan
iv. Chinese Academy of Sciences, Beijing, China
v. Irish Centre for High-End Computing (ICHEC), Galway, Ireland
vi. Leeds University Business School, University of Leeds, UK
vii. Japan Meteorological Corporation, Japan
viii. National Institute of Advanced Studies, India
ix. Postgraduate Institute of Medical Education and Research (PGMER), India
x. Indian Institute of Tropical Meteorology, India

Teaching Experience:

Sep 2024- Present	• Environmental Pollution • Basic of Statistics & Computer Applications	PG Students Dept of Environmental Science, Berhampur University, India

References:

Dr. Saroj Kumar Sahu, Associate Professor, Berhampur University
Email: sks.evs@buodisha.edu.in

Dr. Atul Kumar Srivastav, Scientist E, Indian Institute of Tropical Meteorology, Delhi
Email: atul@tropmet.res.in

Dr. Karn Vohra, Research Fellow, University of Birmingham
Email: k.vohra@bham.ac.uk